Ward Design, Coercion and Aggression in Psychiatric (Intensive) Care

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1 Introduction

In psychiatry, discussions about the ward environment have often focused on Psychiatric Intensive Care Units (PICUs) (Baker et al., 1959; Crain & Jordan, 1979; Goldney et al., 1985; Gove & Lubach, 1969; Higgs, 1970; Mounsey, 1979). In general medicine, literature has demonstrated an association between the ward environment and (mental) health (Gross et al., 1998; Wilson et al., 1983). Studies in this area, including those by Dijkstra et al. (2006) and Watts and Wilson (2009), have connected factors such as hygiene, logistics, design, lighting, colour, accessibility, atmosphere and ambience to the concept of a healing environment.

Evidence of the association between ward design and mental health is, however, limited. This chapter provides a historical overview of research and practical experience into the impact of the ward environment on mental health, aggression, and

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coercive measures in psychiatry. After describing the main developments of Psychiatric Intensive Care Units within mental health, we present some general findings of studies into healing environments, both outside and within psychiatry. In many countries, PICU guidelines first concern a methodical approach to applying evidence-based nursing and team interventions and second, more recently, evidence-based ward design.

In several countries, building guidelines have been developed over the last 20 years. When we look at the development of PICUs, we observe several developments. The first descriptions of PICUs cover initiatives in the United States (Baker et al., 1959; Crain & Jordan, 1979; Gove & Lubach, 1969), England (Basson & Woodside, 1981; Dix & Williams, 1996; Mounsey, 1979), Ireland (Patton, 2013; Raaj et al., 2023), Canada (Musisi et al., 1989), New Zealand (Corey et al., 1986) and Australia (Goldney et al., 1985). In more recent years, PICUs have been established in Norway, Sweden (Faerden et al., 2023), Denmark (Simonsen & Duff, 2020a, 2020b) and the Netherlands (van der Schaaf et al., 2013; Voskes et al., 2021; Vruwink et al., 2012). Most of these countries have or are currently developing PICU guidelines. Many recent guidelines are based on a similar body of evidence. A report for the European Commission (Pincus, 2019) identified PICUs in 14 countries—Australia, Canada, the Czech Republic, Ireland, Wales, Japan, the Netherlands, New Zealand, Switzerland, Sweden, the UK and the USA. As well as these countries, we recently identified PICUs in Denmark (Berring et al., 2023) and Belgium (Zorgnet-Icuro, 2023).

2 History of Psychiatric Intensive Care

One of the first publications covering the concept of a Psychiatric Intensive Care Unit (PICU) concerns a working group document of the World Health Organization (WHO) expert committee on mental health (Baker et al., 1959), which defined several design criteria for psychiatric wards and facilities. This report describes one of the earliest conceptual designs for intensive care units in psychiatry. The criteria were based on expert opinion and included suggestions for single rooms, ward overview, ward design, lighting, colour, location of team rooms and ward accessibility. Ward design varied depending on its function and target population, whether for outpatients, youth, adults, acute admissions or the elderly. Important suggestions of the working group were to organise team rooms and ward management offices near the ward and to divide wards into smaller units with different treatment intensities. However, even though these suggestions were made in the late 1950s, many of the suggestions took decades to be carried out in mental health care.

Gove and Lubach (1969) provided the first evaluation of treatment in a Psychiatric Intensive Care Unit (PICU). This study described the establishment of a stepped-care intensive treatment program for acutely admitted service users of a single county in the west of the USA. The pilot program included all 258 consecutive admitted service users between December 1962 and December 1964. All 171 service users admitted to the same ward a year before were treated as the control group.

The hospital stay for the control group was two and a half times longer than for those in the program, yet no differences in patient characteristics were observed.

Historically, the development of PICUs can be observed to progress in three phases. During the 1960s, 1970s and 1980s, psychiatric intensive care was a methodical approach detailed in various publications, including those mentioned previously. Most of the publications were policy evaluation documents providing limited underlying evidence. Apart from ward size, no other ward-level variables were yet included in any of these reports. This approach mirrored that of intensive care in general hospitals, focusing on specialised care delivered by highly trained professionals, guided by various protocols. However, evidence for this approach was limited because its integration into a rigorous randomised-controlled trial design would have been challenging. Findings were at best based on prospective observational data as in the study by Musisi et al. (1989), but mostly concerned retrospective findings with many shortcomings (Bowers et al., 2008).

Crain and Jordan (1979) published an empirical evaluation of an initiative concerning a new ward environment of eight beds, combined with a stepped-care treatment approach. Tied into each step was a specific nursing care plan, which defined the extent of supervision, privacy, limit-setting techniques and privileges, as well as a set of activities deemed safe for the assessed risk. As many features of the ward milieu were incorporated into the steps, this system began to resemble a contingency hierarchy that progressively reinforced service users' behaviour in the direction of non-destructiveness. In practice, the service user would gain increasing token economy benefits as they advanced from lower to higher steps. The ward was equipped with ample space and also had a high staff-to-patient ratio of above one staff member per patient. The study showed that this separate unit led to fewer violent incidents and shorter hospital stays. It was also associated with fewer staff injuries. The main conclusion of the study was that the PICU did not completely prevent violent incidents. It did, however, 'provide a more humane treatment setting for [violent] individuals whose behavior ordinarily would provoke angry, punitive responses from the environment' (Crain & Jordan, 1979, p. 197).

A later study was conducted in Glenside Hospital, Adelaide, Australia, involving a 3-year observational study in an eight-bed PICU, with six beds located in single rooms (Goldney et al., 1985). Stay at the ward was short at a mean of 5 to 6 days. Approximately 60% of the service users were admitted with psychosis and received haloperidol as treatment. Contrary to previous research by Crain and Jordan (1979), Mounsey (1979), and Basson and Woodside (1981), the hospital stay in the Australian ward was much shorter and only provided the function of crisis intervention and stabilisation through a high-dosage medication protocol.

A study by Musisi et al. (1989) described a PICU in a rural psychiatric hospital. This was a six-bed ward, with a large central room, a television and a conference room. The ward had a nursing desk near the entrance with an overview of all single rooms. The introduction of the ward in 1984 led to a decrease in staff and service users' accidents, a decrease in constant observation and seclusion hours, and a decrease in the number of nursing hours lost to injuries at work. The PICU was appreciated by nursing staff who preferred to work in its more consistent and

controlled environment. In addition, it was felt that the ward environment in other parts of the hospital became more therapeutic. The authors, therefore, concluded that PICUs along such a design were a useful addition to psychiatric settings with important cost and care implications. In effect, this study was the first one to show some empirical benefits of PICUs.

In the 1990s, many of the protocols described by the earlier studies were included in the development of PICU guidelines. From this time onwards, the PICU approach was increasingly confined to small wards, which were partitioned out of larger admission wards and accompanied by building criteria as depicted in the publication by Dix and Williams (1996). They suggested the ward should be positioned at a central location on the ground floor and suggested several security measures including electronic keys, lockable doors and monitoring devices. A clear view of staff over the ward is essential and, therefore, small corridors and dark corners should be avoided in the design. The patient–staff ratio on these wards should be higher, while the nurses should be trained in de-escalation, handling aggression and other techniques to prevent the use of coercive measures.

Moreover, the ward should have practical and recreational facilities, such as sports facilities and a ward telephone. Plants and accessories can be added to create a homely environment. The ward should be designed in a way that facilitates one-to-one guidance and close observation and has between four and eight single bedrooms, aligned along several central rooms.

Subsequently, an evidence-based body of knowledge developed, accompanied by guidelines in several countries between 2002 and the present day. The guidelines include evidence-based nursing and team interventions alongside adjustments to building facilities informed by architectural and social psychological studies of the healing environment. The healing environment, as developed in general hospital care, identifies elements such as space, overview, ward structure, lighting, colour and ward design. The application of healing environment principles in the design of many PICUs varied depending on the ward's location within the hospital design.

3 Healing Environment: A General Hospital Concept Adjusted for Psychiatry

A healing environment refers to a physical setting as well as the organisational culture that supports patients or service users in their recovery (DuBose et al., 2018). As stated, much work has been undertaken examining the effects of the healing environment on patient outcomes in general hospital settings. A study by Stichler (2001) showed that patients in critical care wards of general hospitals have a shorter length of stay, take less pain medication and have fewer notations of disordered behaviour (such as violence) in nursing reports, than those in a less welcoming environment. The following domains were identified by Stichler (2001) as being associated with better outcomes:

1. Ward design.

2. Light, colour, furnishing, sound and privacy.
3. Room design and furbishing as well as an optimal number of rooms.
4. Visibility of all parts of the ward.
5. Family space.
6. Nursing station design and staff support areas.
7. Unit support areas.
8. A circular, triangular or rectangular design.

Dijkstra et al. (2006) published a systematic review of the healing environment in the general healthcare setting. Outcomes predominately concerned clinical recovery and social and psychological well-being. Predominantly positive effects were found for sunlight, windows, odour and seating arrangements. Inconsistent effects were found for sound, nature, spatial layout, television and multiple stimuli interventions. The authors identified important architectural factors: ward layout and placing, and ambient features such as sunlight, sound, odour, window size and spatial design. In addition to these immutable characteristics, interior design features such as plants, seating provisions, television facilities, music facilities and ward colouring were also evaluated. Predominantly positive effects were found for sunlight, windows, odour and seating arrangements. Inconsistent effects were found for sound, nature, spatial layout, presence of a television and multiple stimuli interventions. In general, both the size and direction of effects seem highly dependent on the characteristics of service user populations and healthcare settings.

Similarly, Watts and Wilson (2009) published a review on evidence-based design in paediatric hospitals, examining the impact of physical elements on clinical, psychological and patient/family perceptions. They identified eight studies, published between 1980 and 2008, which found that private space, small ward size, single-room design for intensive care units, specially designed gardens, play centres and nature art were associated with benefits including reducing infections, promoting engagement and supporting coping with illness, particularly for adolescents and their parents.

The concept of a healing environment in psychiatry was first identified and conceptualised from both an architectural and a psychosocial viewpoint within a psychiatric context by Gross et al. (1998). In a large psychiatric hospital in Israel, ward design (along with healing environment principles) was developed by a multidisciplinary team of architects, psychiatrists, other mental health professionals, administrators and service users. The design of inpatient wards in this centre followed several principles of healing environment guidelines for ward design. Service users were not overcrowded or confined to overly small spaces, thereby avoiding forced interaction with too many people. Service users had the opportunity to retreat to their own, single rooms when feeling threatened, and to form beneficial relationships with staff or other patients of their own choice as the ward provided enough space to do so. This was achieved by providing a variety of spaces that supported social interaction, such as a large living room, a dining room with adequate lighting and ventilation, and a spacious lobby and corridors, with residential furniture rather than commonly used institutional pieces. The open porches of this one-story

building showed a gradual transition from the ward to the garden, thus providing further spaces for social interaction, and creating a free and unthreatening atmosphere while at the same time helping to define the environment.

For the design of the ward by Gross et al. (1998), several findings from previous studies were included in the design. In previous years, empirical studies evaluating the impact of ward design revealed various effects of the physical environment on behaviour. Higgs (1970), for example, found a decrease in pathological behaviour after moving service users with schizophrenia to a newly rebuilt, larger unit that featured a comprehensive ward overview and single rooms. Another architectural change that reportedly might bring clinical benefits is the structural division of the ward into smaller units. Wilson et al. (1983) showed the division of wards into small units increased the risk of aggression and coercion, due to less overview and diminished opportunity of keeping enough eye on the patients. Corey et al. (1986) showed even simple redecorations in psychiatric wards, such as changes in furniture style, floor covering and colour scheme, could have a favourable impact on staff and service user perceptions of their physical environment.

These experiences contributed to the concept of a healing environment, which suggests that the ward environment can make a difference in the recovery of the service user. An important body of literature from the 1970s to the 1990s demonstrated that service users have a better outcome in an environment with natural light, elements of nature, soothing colours, meaningful and varying stimuli, peaceful sounds, pleasant views and a sense of beauty (Berring et al., 2023).

Lundin (2021) published a discussion paper presenting a theoretical framework identifying why a healing environment could contribute to increased safety at psychiatric wards. Ulrich et al. (2018) developed this idea into a theoretical model, describing the association between psychiatric disorder, ward design and service user outcome (Fig. 1). The main theoretical point presented in this model is that reduced stress will reduce the risk of negative emotions and thereby reduce aggressive behaviour (Ulrich, 1991; Ulrich et al., 2018).

This stress-reducing theory is possibly the most widely used and accepted theory when it comes to explaining the mechanism between physical interventions, ward environment and health outcomes in general. Ulrich et al. (2018) suggest that when a ward has more stress-reducing factors, coercive measures such as forced medication and physical restraints occur less. The impact of the building environment on ward interaction is complex (van der Schaaf et al., 2013). Collaboration among staff, service users, experts and architects is crucial to address various concerns. Balancing the preferences of all stakeholders can create a healing and secure environment. However, conflicting needs may arise between safety and a healing environment, potentially affecting feelings of safety. While some environments may enhance physical safety, they may compromise the sense of safety for either staff or service users. Staff safety encompasses measures to prevent physical or mental harm during interactions. Improvements in the environment aimed at enhancing staff safety may not simultaneously benefit user safety. For instance, creating command areas can make staff feel more secure by establishing control zones, yet this might reduce patients' sense of safety by introducing a perceived barrier or distance

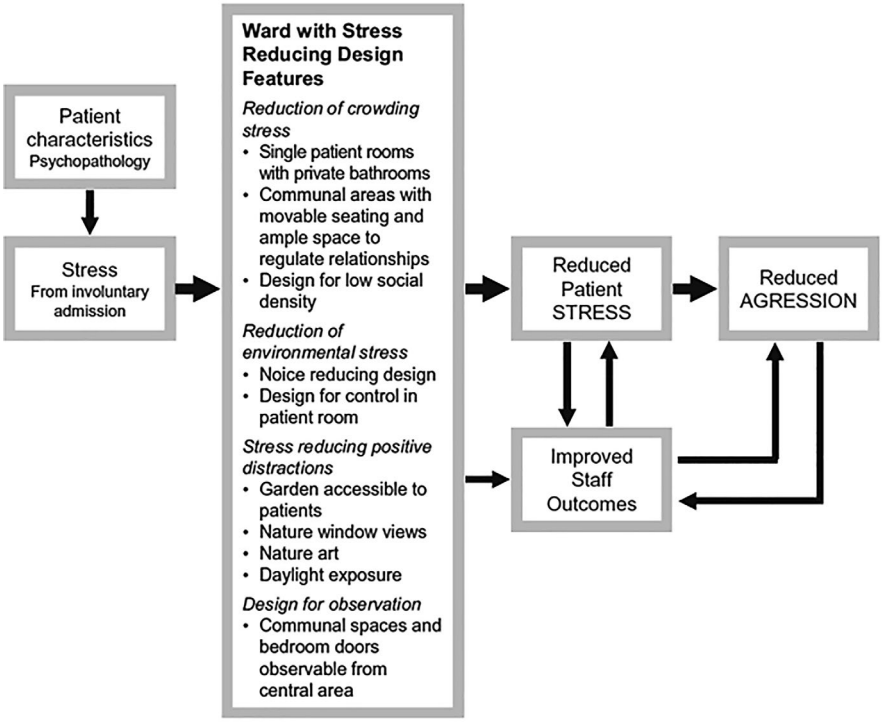

Fig. 1 Theoretical model of ward design (Ulrich et al., 2018). 'Reprinted from Journal of Environmental Psychology, Vol. 57, Author(s), Psychiatric ward design can reduce aggressive behavior, page 14, 2018, with permission from Elsevier'

between them and the staff. Effective communication and spatial planning are essential for mitigating safety threats within the ward (Ulrich et al., 2018).

The threat of aggression is associated with the way nurses deal with service users, but also how the ward environment supports nurses in maintaining safety. Service user safety aims to prevent suicide, self-harm and harm against others (Svensson, 2022; Ulrich et al., 2018). It implies close observation of staff. It implies stepped care of increasing intensive treatment and close observation in different phases of dealing with the service user (Svensson, 2022). Service users should not have the opportunity to plan or act impulsively, resulting in suicide or self-harm. Civil protection aims to prevent service users, who are considered a danger to themselves and others, escape and becoming a threat to society (Svensson, 2022).

The service user's desire for a healing perspective could contradict the safety concerns of the staff or the service user's family (Svensson, 2022). One of the recurring issues concerns the placement of the bed in the services user's room (van der Schaaf et al., 2013). Staff may prefer to see the service user immediately when opening the door; it gives them a feeling of safety to know where the service user is. Additionally, staff want to avoid having to go further into the room, which might

disturb the service user. However, from the service user's perspective, it is not certain that they want to be directly observed. A bed placement by staff requests could imply that the service user's head usually ends towards the façade wall but hinders looking out the window.

Karlin and Zeiss (2006) reviewed publications identifying environmental and therapeutic issues in psychiatric hospital design at a single-ward level. They identified five categories: ambient features, architectural features, interior design features, social features and specific issues related to the service users in care. Ambient features concern lighting, air quality, space and noise. Architectural features are about the constant characteristics of the building. The layout, size and shape of wards, and single or shared rooms promote or hinder participation in treatment activities. Interior design features are about the non-permanent features of a ward. This concerns features as placing of the reception area, the nursing office, the dining room and other common facilities. These features, together with the design and furnishing of service users' rooms, constitute a certain atmosphere which can contribute to a feeling of being at home.

The concept of a healing environment was a major issue in a large nationwide observational Dutch study into the association of psychiatric ward characteristics to the use of seclusion, which may be seen as a proxy for aggressive outbursts (van der Schaaf et al., 2013). It examined seclusion incidents and ward design in 199 wards over 12 months, gathering data from 77 hospitals and a benchmark study. Analysis showed certain design features increased seclusion risk, like limited outdoor space and high service user density, while others, like more private space and comfort, reduced it. The study emphasised the importance of a healing environment, suggesting that intrusive safety measures should be balanced with creating a homelike atmosphere to improve service user well-being and reduce seclusion and restraint needs. Several of the findings of the study were included in the Dutch national guidelines for the High and Intensive Care (HIC) model (Voskes et al., 2021). The guideline included items such as individual rooms and personal bathrooms, comfort rooms, diversity of shared spaces, outside space, a family room, a time-out room, an open nursing front desk, and the use of single intensive care units and a high-security room for emergency interventions. At a national level, these items are seen as contributing to the quality of care. Therefore, they are also considered in the inspections of the Dutch Mental Health Inspectorate.

Variations in ward features within a ward are important for a healing environment in psychiatry (van der Schaaf et al., 2013). Consideration of colours may be important with some authors suggesting incorporating neutral colours in the interior; different colour combinations can have varying effects on mood (Dalke et al., 2006). Brighter colours may be necessary for service users with depression or cognitive disorders (Dalke et al., 2006). Warm blue tones are known to have a soothing or sedating effect (Dalke et al., 2006). Also, closely related colours of the same value and intensity have been reported to have a calming effect. Blue-green colours can harm the mood of service users with depression and less energy. Seclusion room walls should be a calm, but definitive colour, not white or grey (Dalke et al., 2006).

In short, while many publications discuss environment as important in (mental) health care, few studies present real evidence of the effect of ward design on service user outcomes, including symptoms, well-being or recovery, or disordered behaviour, violence or coercion (van der Schaaf et al., 2013; van Melle et al., 2020; Voskes et al., 2021; Vruwink et al., 2012). Most of the studies about healing environments described in this chapter have been small, covering single wards. The studies within the reviews that we have identified in general and psychiatric settings (Dijkstra et al., 2006; Karlin & Zeiss, 2006; Watts & Wilson, 2009), combined many ward features. However, none covered more than one hospital. Evidence was obtained from the combination of data from the different review articles, but not confirmed by studies, combining or pooling any data.

4 Ward Design and Aggression

Around the turn of the twenty-first century, Henk Nijman, a leading researcher in aggression and psychiatry, published two small studies on the association between aggression and ward environment. The first investigated the incidence of aggressive incidents at an acute psychiatric admissions ward in the Netherlands (Nijman et al., 1997). It tested the reliability of the Staff Observation and Aggression Scale–revised (SOAS-r). Importantly, this was the first study looking into the location of incidents within the ward. They found that 48.6% of the incidents took place in day rooms, 19.5% near the staff offices, 14.6% near the outside door of the ward, 7.3% in service users' bedrooms and 6.1% outside the ward, while in 3.9% of incident reports that the location was not specified. The study also identified that the incidents at the ward's main entrance were less severe than those in service user's rooms. In a second study performed in a ward undergoing reconstruction, a modest association between crowding, aggression and ward occupation ratio was identified (Nijman & Rector, 1999). No association, however, between ward design and aggression was found.

Sato et al. (2017) investigated the association between ward characteristics and aggression in 30 wards across 20 Japanese psychiatric hospitals. Over 6 months, they collected violence data using the Japanese version of the Staff Observation Aggression Scale—revised (SOAS-r), during which 443 incidents were reported. The study differentiated between three types of psychiatric wards: emergency wards for severely ill patients for short stays, featuring spacious environments, higher patient-staff ratios and better-trained staff, probably comparable to a PICU; acute wards for less severely ill patients, also for short stays; and standard wards for longer-term treatment aimed at full recovery. Aggression was mostly directed at staff in acute and emergency wards, with medication enforcement being a leading provocation in acute wards and verbal aggression the most common method. Conversely, in standard wards, aggression typically involved other service users, with physical means used, leading to reported pain and the application of seclusion. The findings underline the ward environment's significant role in influencing aggressive behaviour, with acute wards experiencing the highest aggression rates, followed by

emergency and standard wards, respectively. They did not find any difference in the time of day that incidents occurred but did find that the characteristics of aggressive incidents are shaped by the clinical condition of patients, their interactive relationships with staff and the ward environment, which includes staff assignments.

Similarly, Vaaler et al. (2006) compared the number of aggressive incidents in a PICU to those in a regular admission ward. The PICU was a small unit of four single rooms, a sitting room, a dining room and a staff room providing a full ward overview through one-way screens at the centre of the ward. This study showed that in the PICU, there was a lower risk of aggression, as assessed by the Brøset Violence Checklist (BVC), and fewer actual aggression incidents, as measured with the SOAS-r. Even after adjusting for service user characteristics and the severity of psychosis, the difference remained significant.

Weltens et al. (2021) conducted a systematic review of the association between aggression and service user, staff and ward factors. The prevalence of aggressive behaviour in psychiatric wards varied substantially (8–76%). Service user risk factors were diagnosis of psychotic disorder or bipolar disorder, substance abuse, a history of aggression and younger age. Staff risk factors included male gender, unqualified or temporary staff, job strain, dissatisfaction with the job or management, burn-out and quality of the interaction between service users and staff. Staff protective factors were a good functioning team, good leadership and being involved in treatment decisions. Significant ward risk factors were higher bed occupancy, busy places on the ward, walking rounds, an unsafe environment, a restrictive environment, lack of structure during the day, smoking and lack of privacy.

In short, the studies of Nijman (Nijman et al., 1997; Nijman & Rector, 1999) suggest that aggression is often determined by interactions between staff and users. Most of the aggression occurred at the locations where staff and users come together, such as day rooms and staff offices. This is supported by many studies that have shown that staff–patient interaction is the most frequent precursor of patient violence, as identified in a large literature review by Bowers et al. (2011). Crowding and ward occupation rates have shown some association with aggression, while ward design has shown none. Sato et al. (2017) found that ward type was a predictor of the frequency and severity of aggression, with acute psychiatry wards showing the highest rates, followed by emergency wards—like PICUs—and the lowest in standard wards. In acute and emergency wards, staff were primarily the targets of aggression, whereas in standard wards, other patients were more often targeted. Similarly, Vaaler et al. (2006) found that a PICU experienced less aggression compared to a regular admission ward, even after accounting for patient characteristics. The review by Weltens et al. (2021) supported these findings indicating that factors such as high bed occupancy, crowded areas, an unsafe and restricted environment with limited overview, lack of structure and privacy issues were linked to increased aggression.

5 Ward Design Processes in Clinical Practice

To illustrate the process of redesigning an acute admission ward following the guiding principles of a healing environment and current knowledge about ward design, we give an example of the renovation of the High Intensive Care ward at Amsterdam University Medical Center in the Netherlands. The ward of interest is a 12-bed closed psychiatric admission ward in a university hospital. Service users admitted to the ward are adults with various psychiatric diagnoses, with potential somatic comorbidity. Most admissions are acute and involuntary. Approximately one-quarter of admissions are tertiary referral admissions, the majority are general psychiatric admissions from the municipal psychiatric crisis facilities. The original (pre-renovation) ward consisted of two separate units with six single-service user rooms with private bathrooms. Both units had a multifunctional room for (primarily) private conversations or activities, a living room, kitchen, dining room and smoking room. Both units had access to an enclosed garden and to a large nursing station. Next to (but outside), the ward was a laundry room, four seclusion rooms, offices of the ward manager and secretary, a meeting room and a room designed for occupational therapy.

This ward was renovated in 2019 after years of preparation. Part of the preparation was the implementation of the High Intensive Care model (Voskes et al., 2021) and changing (nursing) staff's attitudes towards ward safety and the use of coercive measures using regular training and supervision. The mission of the future ward design was to support and facilitate the wards' ambition to improve the recovery-oriented environment, intensify close contact between service users and staff members and reduce coercive measures. Specifically, the ward required more facilities for intensive (one-to-one) nursing care without using coercive measures and more space to prevent overcrowding. Coincidentally, the hospital would become smoking-free (inside and outside) during the renovation, so the smoking rooms would not return afterwards.

To create a more spatial ward design, the concept of two separate units was abandoned, and a larger, 12-bed ward was created. Furthermore, the corridor where the seclusion rooms, offices and occupational therapy room were situated was included in the ward. As a result, the size of the ward increased by (approximately) 30%, and we substantially improved the possibilities of service users to, e.g., do their laundry or go to the therapy room without needing staff members to unlock doors for them. The ward consisted of three different areas. These areas were not separated by walls or doors but became recognisable by different colours on the wall. Two of these areas were considered 'living areas'. The service user's rooms, comfort rooms, living rooms, dining rooms, kitchens and sports rooms were situated here. The third is the 'working area'. The conversation rooms, therapy rooms, laundry room, front desk and intensive care units were situated here. By dividing living areas and working areas, we aimed to create a safe space for service users where no intense therapy, meetings or interventions were performed. Although exceptions were possible, this remains a guiding principle for care planning at the ward.

The environment was designed to encourage interaction and engagement but to prevent moments of potential conflict between service users. Rooms and corridors were designed so that service users could always avoid a confrontation with another service user. This meant a minimum of corners and more than one entrance/exit in common areas. Rooms were divided with large closets that do not reach the ceiling, also to establish a feeling of openness for those present in the ward. When designing the ward, we used lots of wood and warm, natural colours. When choosing materials and furniture, we maintained a delicate balance between hospitality and aesthetics on the one hand and molest-proof and hygiene on the other. Materials and furniture needed to be strong and easy to clean, but not have a molest-proof, institutional look. When making choices about floors, ceilings and curtains, we chose materials that could improve the acoustics of the ward. The ward implemented several acoustic panels on the wall with a comforting print to improve the sound even more. Lights had warm colours and could be switched off during the night (contrary to the old situation). Furthermore, lighting in the ward was evenly divided in the room by large ceiling lamps. The old situation with small, bright lights resulted in 'light stains' on the ceiling, resulting in a disharmonious picture of the environment.

The nursing station was transformed into an open front desk, which is the heart of the work area and basically, the heart of the ward. The front desk contained a workstation and telephone, but nurses usually work throughout the ward with mobile phones and computers on wheels (COWs). Behind the front desk, locked closets contained all systems that were previously situated in the nursing station, such as telephones, walkie-talkies, light buttons and fire extinguishers. A large round table was placed in the space in front of the front desk to create a hospitable reception area. This has proven to be a popular meeting area for service users and staff members. Also, for hospitality purposes, this area contained a panel with names, professions and pictures of the wards' medical, nursing, therapeutic and supporting staff members. Care workers created a book with personal information for service users to get to know them to improve the feeling of equality between service users and staff members (based on Safewards 'Know each other' intervention) and a guestbook for service to leave hopeful messages at discharge for current or future service users (based on Safewards 'Goodbye messages' intervention).

The corridor next to the work area contained three small conversation rooms, one large meeting room, a laundry room for service users and a room for occupational therapy. Furthermore, two intensive care units (ICUs) were situated in this area. The ICUs were designed to provide intensive nursing care to individual service users in a private area when interaction with other service users counteracts their recovery process. The ICUs contained a bedroom, a small living room with a television and a bathroom. Both units also had access to a private garden. One of the ICUs had room for a next-of-kin to stay overnight if necessary. One of the ICUs was equipped with a high-security room (HSR) in which seclusion is possible in case of an emergency. The HSR had a large window, so the secluded service user could always see the staff member on duty. Furthermore, the HSR contained a toilet, sink, bed and multimedia system. The service user secluded in the HSR could use the multimedia system to operate several aspects of the room, such as light, colours, music, curtains

and the window that can be opened for fresh air. This system is important to give a service user in seclusion a minimum amount of autonomy. The personal belongings of a service user in seclusion can be stored in a closet with a transparent front, which is visible from the window of the HSR. This can prevent conflict about the security of personal belongings between service users and staff members. Other than the HSR, none of the doors inside the ward can be closed from the outside. Not every room is accessible for all, but once inside, one can freely exit any room.

In the living areas, service users can use the living room with a television and a decently equipped kitchen. The cooker and stove can be switched off by staff members when specific service users may cause dangerous situations using them. The comfort rooms also had a television and an electric fireplace to induce a warm atmosphere, together with a couch and relax-seat. Every service user had a single bedroom with an en suite bathroom. The rooms had a bed that could be moved, a closet and a table and chair. The rooms and bathrooms were built to prevent the opportunity for hanging, such as acclivous doorknobs, shower heads and flexible coat racks. The floor had two different colours as a natural boundary for staff members to wait by the door until they were invited to come in. Every service user had a patient badge to enter their room whenever they wanted.

With these and many other design features, we aimed to combine a healing environment with the environmental safety precautions that are needed at a closed ward for acute psychiatric admissions. Some staff members were sceptical towards not having a nursing station and the abandonment of closed rooms (in the old situations, the ward had access to 4 seclusion rooms, and 4 of our 12 service user's rooms could be used to seclude a service user in its room). However, the warm and hospitable environment had an effect not only on service users but also on staff members. Service users responded on numerous occasions that they felt like they checked into a hotel, despite their mostly coerced situation. While psychiatric wards tend to deteriorate faster than other professional environments and vandalism does still occur occasionally, warm surroundings stimulate staff members and service users to keep the environment clean and intact. We believe that the ward design now supports and stimulates our perspective on high-quality, intensive psychiatric care, which could only be accomplished by a long and intense collaboration between service users, staff members, administrators, architects and contractors to achieve such goals.

6 Summary

This chapter has offered an overview of the knowledge regarding the impact of the ward environment on service users, including the incidence of aggression and the application of seclusion and restraint. Since the 1970s, the physical ward environment has been incorporated into mental health policy. Especially over the last two decades, studies have provided evidence of the association between ward environment, safety and the risk of aggression or the use of coercive measures. Several concepts are important in this respect: the healing environment, the establishment of PICUs and the identification of the ward environment as a risk factor for aggression.

For each specific situation, tailor-made advice is needed on how to ensure a safe ward through adjustments to the ward environment and, potentially, its location.

The healing environment relates to many aspects including the ward overview, colour, lighting, the use of plants at the ward and the proximity of a garden or outdoor space. Regarding PICUs, there is a broad consensus that small wards, staffed by highly trained nurses and maintaining a high patient-to-staff ratio are important during the acute phase of the psychiatric illness. The findings of studies investigating the relationship between ward environment and aggression, as well as the use of seclusion and restraint, align with many of the recommendations made in the healing literature. Overall, replacing the classic nursing station with an open front desk and preventing overcrowding are considered key factors in enhancing safety. Additionally, room design and bed placement contribute to increased safety. In designing wards, it is advised to engage service users and experiential experts along with staff in creating a safe environment for the care of those in the acute phase of their illness.

Concerning the direct association between ward environment and aggression, only a few quite small studies could be found covering an investigation of ward design and aggression. The findings of these studies are corroborated by a large study about coercion, which may be seen as a reasonably good proxy for aggression. The first investigated the association between ward design and seclusion and identified 14 design features of wards that had a significant effect on the risk of being secluded (van der Schaaf et al., 2013).

7 Recommendations

From all these studies as well as from practical experience, we can identify several building environment recommendations associated with improving well-being and reducing aggression or coercive measures. Reasoning from outwards inside the first is the placing of the ward within its environment. Preferably a psychiatric (intensive care) ward should be on the ground or on the first floor. It should be not completely incorporated into a general hospital but placed in a separate building with a limited number of beds. It should be in an environment providing direct access to nature. The ward itself should preferably have enough common space to allow a good overview of each of the service users' rooms. The rooms should be for one service user. The interior of the rooms should be adjusted to the service user's preference, except when the service user's behaviour cannot be relied upon. In that case, the placing of the bed needs to be done in a way the space around the bed is enough to move around, both for the professional and the service user. For all these spaces, factors such as light, colour, furnishing and size all contribute to less aggression, less coercion inversely and more safety. Preferably ward redesigning should occur in close collaboration with service users. Whenever psychiatric intensive care wards are meant to move or be refurbished or even rebuilt these recommendations could be considered. Supervision by an expert in the domain of healing environment is

crucial, to avoid making errors in the building environment, as these errors can be very expensive and may consume an important portion of a hospital's operational capacity.

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